Square abscissas and direct charts for the elliptic subgroup

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Abstract

Under the stated elliptic group hypothesis, two square equations describe exactly the nonzero points of the cyclic subgroup used in the arithmetic construction. No parity condition on the index is necessary. A cubic change of coordinates gives one rational quartic presentation that covers every point, with no choice among component charts.

1 The subgroup described by a square abscissa

Put

$$K = \mathbb{Q}(\sqrt{2}), \qquad d = 3 - \sqrt{2}, \qquad D = d^2 = 11 - 6\sqrt{2}, \qquad E : y^2 = x^3 - Dx.$$

Let O denote the identity of E, and set

$$R_0 = (1 + \sqrt{2}, 2\sqrt{2}), \qquad P = 2R_0.$$

The arithmetic hypothesis in this note is

$$2E(K) = \mathbb{Z}P. \tag{1}$$

We also assume, as in the source construction, that P has infinite order. Thus every $Q \in E(K)$ has a presentation $Q = mR_0 + T$ with $m \in \mathbb{Z}$ and $T \in E[2](K)$.

Lemma 1. *The map*

$$\delta : E(K) \longrightarrow K^\times/K^{\times 2}, \qquad \delta(O) = 1, \quad \delta((0,0)) = -D, \quad \delta(Q) = x(Q) \quad (x(Q) \neq 0),$$

is a group homomorphism.

Proof. Consider a nonvertical line $y = \lambda x + \mu$. Its intersections with E, counted with multiplicity, have abscissas given by

$$x^3 - \lambda^2 x^2 - (D + 2\lambda\mu)x - \mu^2 = 0.$$

When $\mu \neq 0$, the product of the three abscissas is μ^2. When $\mu = 0$, one intersection is $(0,0)$ and the product of the other two abscissas is $-D$. In either case the product of the three δ-classes is trivial. Vertical lines meet Q, $-Q$, and O, whose classes also have trivial product. These identities, including tangent lines through their intersection multiplicities, give the assertion by the chord and tangent law. □

Theorem 2. *Assume* (1). *The points arising from*

$$x = h^2, \qquad j^2 = h^4 - D, \qquad y = hj, \qquad h, j \in K, \tag{2}$$

are exactly $E(K) \cap (\mathbb{Z}P \setminus \{O\})$. *Every such point has* $h \neq 0$ *and* $j \neq 0$.

Proof. Both real embeddings of d and D are positive. Consequently $h = 0$ would imply $j^2 = -D$, which is impossible. If $j = 0$, then $h^2 = d$ or $h^2 = -d$. The second alternative is excluded by signs. The first is excluded by $N_{K/\mathbb{Q}}(d) = 7$, since the norm of a square in K is a rational square. Thus the point $Q = (h^2, hj)$ is affine and is not a point of order two.

Write $Q = mR_0 + T$ as above. The four two-torsion points and their classes are

T	O	$(0,0)$	$(d,0)$	$(-d,0)$
$\delta(T)$	1	$-D$	d	$-d.$

Every representative in the second row has the same sign at the two real embeddings. In contrast, $\delta(R_0) = 1 + \sqrt{2}$ has opposite signs at those embeddings. Since $\delta(Q)$ is a square class, the homomorphism property forces m to be even. Write $m = 2k$. Then $Q = kP + T$, and $\delta(P) = \delta(2R_0) = 1$ implies $\delta(Q) = \delta(T)$. Of the four listed classes only 1 is a square: $-D$ and $-d$ are negative, while d has nonsquare norm. Hence $T = O$ and $Q = kP$, where $k \neq 0$ because Q is affine.

Conversely, if $Q = kP$ with $k \neq 0$, then $Q = 2(kR_0)$ and $\delta(Q) = 1$. The point is nontorsion, so $x(Q) \neq 0$. Choose $h \in K^\times$ with $h^2 = x(Q)$ and put $j = y(Q)/h$. The equation of E gives $j^2 = h^4 - D$, as required. □

2 A rational quartic presentation

Write $h = h_0 + h_1\sqrt{2}$ and

$$U = h_0^4 + 12h_0^2h_1^2 + 4h_1^4 - 11, \qquad V = 4h_0h_1(h_0^2 + 2h_1^2) + 6.$$

Thus $h^4 - D = U + V\sqrt{2}$. Define

$$a = 2U + V, \qquad b = -(U + V),$$
$$A = 6U^3 - 9UV^2 - 4V^3, \qquad B = -4U^3 + 6UV^2 + 3V^3.$$

Then the square-root equation for j is equivalent to

$$Y^4 - 4AY^2 + 8B^2 = 0, \qquad Y \neq 0. \tag{3}$$

Every such rational root recovers j by

$$j = \frac{a - b\sqrt{2}}{2U^2 - V^2}\left(\frac{Y}{2} + \frac{B}{Y}\sqrt{2}\right). \tag{4}$$

Corollary 3. *Under (1), the formulas above represent all nonzero points of $\mathbb{Z}P$ by two rational base coordinates h_0, h_1 and one finite quartic root Y. Every permitted rational root is simple. No finite choice of chart is required.*

Proof. Theorem 2 reduces the assertion to the square-root presentation. If $j = u + v\sqrt{2} \neq 0$, then $U = u^2 + 2v^2$ and $V = 2uv$, and direct expansion gives

$$(a + b\sqrt{2})j = 2(u^3 - 2v^3) + (4v^3 - u^3)\sqrt{2}.$$

The rational component is nonzero because 2 is not a rational cube. The multiplier has norm $2U^2 - V^2 \neq 0$, since $(U, V) \neq (0, 0)$ and 2 is not a rational square. Its square times j^2 is $A + B\sqrt{2}$. If Y is twice the rational component of $(a + b\sqrt{2})j$, these identities give (3) and (4).

Conversely, a nonzero root Y gives

$$\left(\frac{Y}{2}+\frac{B}{Y}\sqrt{2}\right)^2 = A + B\sqrt{2} = (a+b\sqrt{2})^2(U+V\sqrt{2}),$$

so (4) gives the required square root j. At such a root the derivative of the quartic is

$$4Y^3 - 8AY = 8Y\, N_{K/\mathbb{Q}}\left(\frac{Y}{2}+\frac{B}{Y}\sqrt{2}\right) \neq 0.$$

This proves the simple-root assertion as well. □

The condition $Y \neq 0$ must be retained when this presentation is used: the quartic equation alone does not justify the displayed rational recovery formula at $Y = 0$. In a finite-algebra construction it is imposed by the localization step, with its unit condition checked separately. The general version of this square-root chart, valid for every quadratic extension of a number field, is proved in the companion paper *A single quartic chart for square roots in a quadratic extension.*